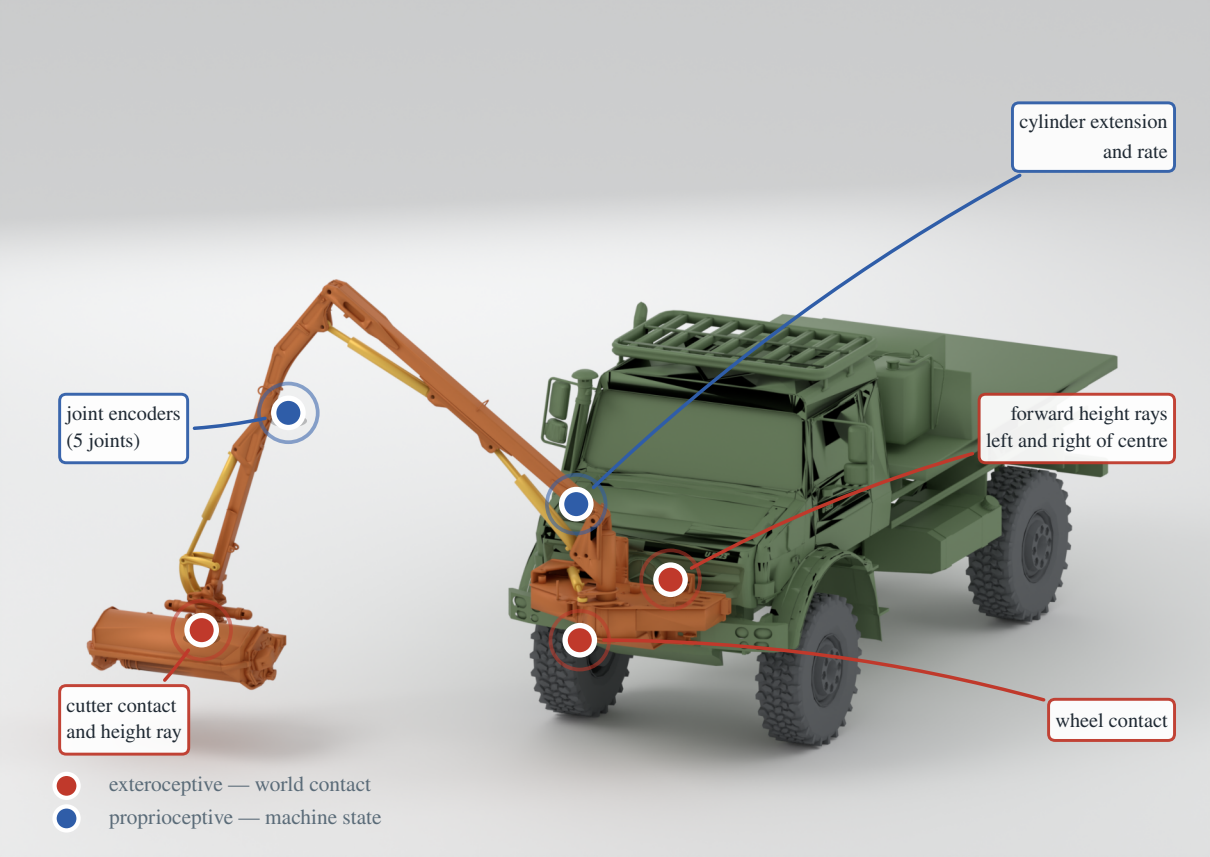

cylinder extension and rate
joint encoders (5 joints)
forward height rays left and right of centre
cutter contact and height ray
wheel contact
exteroceptive — world contact
proprioceptive — machine state